Recall & Practice

Loops • Conditionals • Functions

Module 1: Python Foundations

```python
def evens(nums):
    for n in nums:
        if n % 2 == 0:
            print(n)
evens(range(6))
```

By the End of Today, You'll Be Able To…

1

Trace loop, conditional, and function code cold

2

Spot and explain a planted bug

3

Write a TCEO prompt with one example, then check it

WELCOME BACK

Today’s Format

Recall first, with no safety net. Then one new skill: asking AI for help.

Today: Recall, Then AI Prompting

RETRIEVE

Pure retrieval practice on loops, conditionals, functions.

PACE

Quick-fire rounds, low stakes, high energy.

SUPPORT

Ask questions any time — no pressure.

CHECK FOR UNDERSTANDING

What does `range(3)` produce?

A

0, 1, 2, 3

B

1, 2, 3

C

0, 1, 2

ANSWER KEY

`range()` Recall

Correct answer: 0, 1, 2

`range(3)` starts at 0 and stops before 3 — three values, not the number 3 itself.

CHECK FOR UNDERSTANDING

A variable created inside a function — local or global?

A

Global

B

Local

ANSWER KEY

Local vs. Global Recall

Correct answer: Local

It only exists while that function is running, and nothing outside the function can see it.

QUICK CHALLENGE

Even values in `[[1,2],[3,4],[5,6]]` are added. What's `total`?

A 21

B 12

C 6

ANSWER KEY

Nested Loop + Conditional Trace

Correct answer: 12

The even values are 2, 4, and 6. Add them: 2 + 4 + 6 = 12.

Watch It Run: Adding the Even Values

Step 1 of 6

total starts at 0

```
grid = [[1, 2], [3, 4], [5, 6]]
total = 0
for row in grid:
    for value in row:
        if value % 2 == 0:
            total += value
print(total)
```

OUTPUT

ANIMATED IN CLASS: watch it run in the lecture recording

CHECK FOR UNDERSTANDING

Fill the blank: `if age ____ 18:` means “at least 18.”

A

`>=`

B

`<=`

C

`==`

ANSWER KEY

Comparison Operator Fill-In

Correct answer: >=

“At least” includes the threshold itself and anything above it.

CHECK FOR UNDERSTANDING

`scores.py` holds Python code but starts with `#!/bin/bash`. You run `./scores.py`. What happens?

A Python runs it, since it ends in `.py`

B bash runs it and errors

C `zsh` runs it, since it's your shell

ANSWER KEY

The Shebang Picks the Program

Correct answer: B. bash runs it and errors

`./scores.py` hands the file to the program line 1 names. `#!/bin/bash` picks bash, which can't read Python. `python3 scores.py` would still work.

TURN AND TALK

Put `area(3, 4)` in Execution Order

A

`width` and `height` become 3 and 4

B

the body runs and returns 12

C

`area` is called with 3 and 4

CODE ALONG

Open the Code-Along

Open
`Module1_Lesson5_recall-practice-loops-conditionals-functions_STUDENT.py`
Run it after every section: `python3` + the file name

Sections, in order:
1. Loop + Conditional + Function, Together

Loop + Conditional + Function, Together

python

```python
def count_passing(scores, threshold):
    count = 0
    for score in scores:
        if score >= threshold:
            count += 1
    return count

print(count_passing([55, 82, 91, 40, 76], 70))
```

CODE-ALONG DONE

Back to the Slides: Review

You just wrote it and ran it. Now back to quick-fire recall, without the code in front of you.

QUICK CHALLENGE

`set_total()` creates `total = 100` locally. What does `print(total)` outside do?

A Prints 100

B Prints `None`

C Raises `NameError`

ANSWER KEY

Scope Bug: `NameError`

Correct: Raises a `NameError`

`total` is local to `set_total`. It is created and destroyed inside that call, so outside code cannot reach it.

TIP

“Why Pass the Passing Score In as a Parameter?”

HARDCODED

```
if score >= 70:
```

PARAMETER

```
count_passing(scores, threshold)
```

Answer like an interview, with PREP: point, reason, example, point.

MODEL ANSWER

Model Answer: PREP

POINT

Pass the cutoff in as `threshold`, not hardcoded 70.

REASON

One function then works for any cutoff: 70, 80 or 90.

EXAMPLE

With 80 instead of 70, the same call returns 2, not 3.

POINT, RESTATED

A parameter whenever the rule can change; hardcode only one-offs.

QUICK CHALLENGE

How many values in `range(6)` are even?

A 3

B 2

C 6

ANSWER KEY

Loop + Function Trace

Correct answer: 3

`range(6)` gives 0, 1, 2, 3, 4, 5. The even values are 0, 2, and 4.

CHECK FOR UNDERSTANDING

`range(3)` outside × `range(4)` inside: total inner-body iterations?

A 7

B 12

C 3

ANSWER KEY

Nested Loop Iteration Count

Correct answer: 12

The inner loop runs 4 times for each of 3 outer passes: 3 × 4 = 12.

QUICK CHALLENGE

`average(nums)` uses `total = n` inside the loop. What's the bug?

A Use `len(n)`

B Put `return` inside loop

C Use `total += n`

ANSWER KEY

Spot the Accumulator Bug

Correct: `total = n` should be `total += n`

`total = n` overwrites the running sum every pass instead of accumulating it.

Prompting: How to Ask AI for Help

AI predicts the next words. A clear question gets an answer about your code that you can check.

CHECK FOR UNDERSTANDING

A chatbot answers with confidence. What is it doing?

A Predicting likely next words

B Looking it up in a database

C Running your code to test it

ANSWER KEY

Module 0 Recall: How a Chatbot Answers

Correct answer: Predicting likely next words

It writes a fluent guess, one piece at a time. It can sound sure and still be wrong, so verify before you trust it.

Four Facts That Change How You Prompt

TOKENS

Text is chopped into small pieces called tokens, like words or parts of words.

CONTEXT WINDOW

It only sees what fits in this chat. Long chats push early details out.

TRAINING CUTOFF

It learned from data up to a date. For anything newer, paste in a source.

NON-DETERMINISM

The same prompt can get a different answer. If it changes, check it.

PROMPT SHAPE

Task, Context, Evidence, Output

```python
def set_total():
    total = 100

set_total()
print(total)
```

TASK

What you want done, in one sentence.

`Explain why I get a NameError. Do not fix it.`

PROMPT SHAPE

Task, Context, Evidence, Output

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def set_total():
    total = 100

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print(total)
```

TASK

What you want done, in one sentence.

`Explain why I get a NameError. Do not fix it.`

CONTEXT

What you are working on and already know.

`I am learning scope. total is set inside set_total().`

PROMPT SHAPE

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def set_total():
    total = 100

set_total()
print(total)
```

TASK

What you want done, in one sentence.

```
Explain why I get a NameError. Do not fix it.
```

CONTEXT

What you are working on and already know.

```
I am learning scope. total is set inside set_total().
```

EVIDENCE

Your real code, error, or a few rows.

```
[paste my code]
NameError: name 'total' is not defined
```

PROMPT SHAPE

Task, Context, Evidence, Output

```python
def set_total():
    total = 100

set_total()
print(total)
```

TASK

What you want done, in one sentence.

```
Explain why I get a NameError. Do not fix it.
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CONTEXT

What you are working on and already know.

```
I am learning scope. total is set inside set_total().
```

EVIDENCE

Your real code, error, or a few rows.

```
[paste my code]
NameError: name 'total' is not defined
```

OUTPUT

The answer's shape, and what to avoid.

```
Ask me one question that helps me find it. No code.
```

Show One Example of the Answer You Want

BEFORE

```
Which of these
shifts are long?

[paste 5 rows]
```

AFTER: ONE EXAMPLE

```
Label each row of this shift log with "yes" if the
volunteer worked more than 4 hours, otherwise "no".
Example: Dev,Wed,5 gives yes, because 5 is more than 4.
Now label the other 5 rows:
Amara,Mon,3
Amara,Tue,4
Priya,Thu,6
Luis,Fri,4
Dev,Fri,1
```

Then check it. Judge two rows yourself before you trust the rest.

TURN AND TALK

Rewrite "Fix My Code" as a TCEO Prompt

```python
def count_long_shifts(hours):
    count = 0
    for h in hours:
        if h >= 4:
            count += 1
        return count

print(count_long_shifts([3, 4, 6, 2, 5]))
```

IN PAIRS: 10 MINUTES

1. Rewrite "fix my code" as Task, Context, Evidence, Output. Add one example of the answer shape.
2. Send the vague prompt and yours to an AI tool. Compare the two answers.
3. Make the fix yourself. Check it: `count_long_shifts([5, 6])` should be 2.

It printed 0. By hand, 3 shifts are 4 hours or more: 4, 6 and 5.

ASKING AI TO REWRITE CODE? For your own projects and practice. For graded work, ask it to explain instead.

- Paste only the lines you want changed, not the whole file.
- Keep a small test file, like `count_long_shifts([5, 6])` gives 2. Run it before and after, to prove the rewrite still gets the right results.

MODEL ANSWER

Model Answer: The Same Bug, as a TCEO Prompt

TASK

```
Help me understand why my function
returns 0. Do not write or fix the code.
```

CONTEXT

```
I am learning loops and functions. It
should count shifts of 4 or more hours.
```

EVIDENCE

```
[paste the code] It printed 0. By hand I
count 3: 4, 6 and 5.
```

OUTPUT

```
Ask me one question, then name one line
to check. Example from another bug: Line
4 uses > but needs >=
```

Your Prompt Checklist

TCEO

All four parts, and what the AI must avoid.

ONE EXAMPLE

Show one answer in the exact shape you want back.

VERIFY

Check it against a case you worked by hand. Your run wins.

So Far, We've Covered…

SOLID

Control flow and functions

NEXT

Big-O intuition